A Low-Power Low-Noise Two-Stage CMOS Operational Amplifier for Audio Signal Processing Applications

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***Abstract*—This paper presents a low-power, low-noise two-stage CMOS operational amplifier targeted for audio signal processing applications. The proposed design utilizes a differential input stage followed by a common-source gain stage. A Miller compensation network is implemented to improve closed-loop stability and transient performance. The proposed op-amp achieves a low input-referred noise of 8.6 nV/$\sqrt{Hz}$ at 1 kHz and maintains good linearity with a total harmonic distortion of 0.00041%. The amplifier operates at a supply voltage of 1.8 V and consumes only 0.92 mW of power. The op-amp achieves a DC gain of 91.4 dB, unity gain bandwidth of 46 MHz and a phase margin of 61°, providing stable operation for closed-loop applications. The amplifier also provides a slew rate of 14 V/μs and a settling time of 91 ns within 0.1% accuracy. The obtained results indicate that the proposed CMOS operational amplifier is suitable for low-voltage audio front-end circuits and analog signal conditioning applications.**

***Index Terms*—CMOS operational amplifier, low noise design, low power design, audio signal processing, analog integrated circuit, two-stage op-amp**

I. Introduction

The low-noise operational amplifier is one of the most important circuits used in analog design. In audio signal processing systems, the quality of the first amplification stage significantly affects the overall performance of the system. Audio signals from sources such as microphones and sensors are typically low amplitude, requiring amplifiers with low input-referred noise and high linearity to preserve signal integrity. In addition, low-power consumption has become increasingly important for portable and battery-operated audio devices.

Conventional operational amplifiers are generally designed for wide-ranging applications and may not provide an optimal balance between noise, power consumption, and performance required for audio front-end circuits. Increasing bandwidth and gain often results in higher power consumption, while aggressive power reduction can degrade parameters such as voltage gain, bandwidth and noise performance.

The objective of this work is to design a low-voltage CMOS operational amplifier that provides low-noise performance and reduced power consumption while maintaining sufficient gain, bandwidth, stability and linearity for audio signal processing applications. The proposed design operates at a supply voltage of 1.8 V and aims to achieve sub-milliwatt power consumption with suitable dynamic performance.

The proposed amplifier is designed and simulated using Cadence Virtuoso and its performance is evaluated in terms of gain, bandwidth, noise, power consumption, transient response and linearity.

II. Proposed Op-Amp Architecture

The proposed operational amplifier is a two-stage CMOS architecture designed to achieve low-power consumption and low-noise operation under a low supply voltage condition. The overall structure consists of a differential input stage, a common-source gain stage, a bias generation circuit and a frequency compensation network. The block diagram of the proposed architecture is shown in Fig. 1.

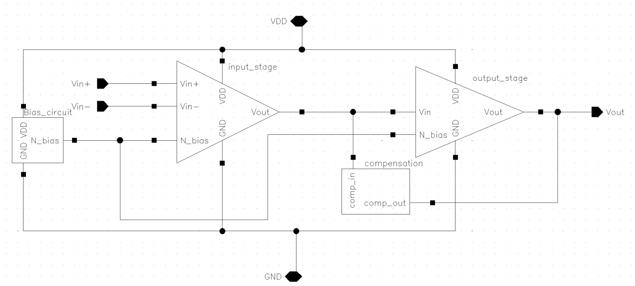

Fig. 1. OP-AMP Architecture

A. Differential Input Stage

The first stage of the operational amplifier consists of an NMOS differential pair with a PMOS active load. This configuration provides differential-to-single-ended conversion while achieving high voltage gain and improved transconductance efficiency. The input stage is responsible for amplifying the input voltage difference and contributes significantly to the overall noise performance of the amplifier. The selection of the input transistor type and bias current is optimized to achieve a balance between input-referred noise and power consumption. The CMOS circuit of the differential pair is shown in Fig. 2.

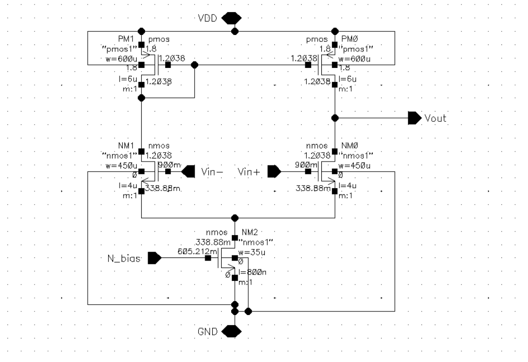

Fig. 2. Differential Pair

B. Second Gain Stage

The second stage is implemented using a PMOS common-source amplifier with an NMOS current sink load. This stage provides additional voltage gain and increases the overall open-loop gain of the operational amplifier. The output stage is designed to maintain sufficient output swing under the 1.8 V supply constraint while providing suitable driving capability for closed-loop applications. The CMOS circuit of the gain stage is shown in Fig. 3.

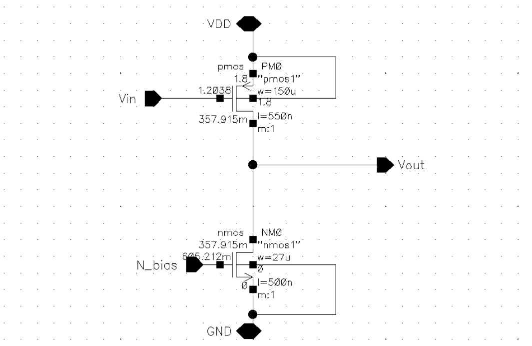

Fig. 3. Common Source Ouput Stage

C. Bias Circuit

A beta multiplier bias circuit is used to generate a stable bias voltage for the current sink transistors. This voltage-based biasing approach provides consistent current operation while maintaining low-power consumption under the 1.8 V supply constraint. The CMOS circuit of the bias generator is shown in Fig. 4.

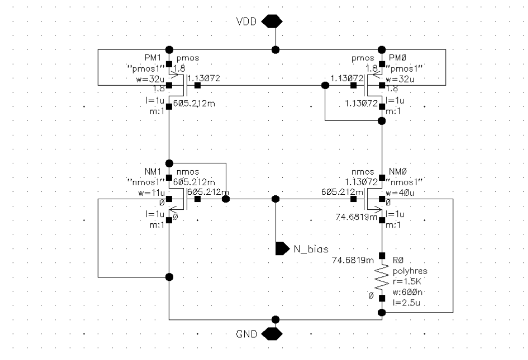

Fig. 4. Bias Circuit

D. Frequency Compensation

To achieve stable closed-loop operation, a Miller compensation network is implemented between the first and second gain stages. The compensation network consists of a compensation capacitor and a nulling resistor. The capacitor provides dominant pole compensation, while the resistor improves phase behavior by reducing the effect of the right-half-plane zero. A compensation capacitor of 3 pF with a series resistance of 417 Ω is used in the proposed design.

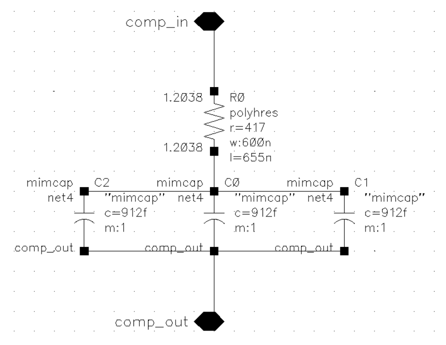

Fig. 5. Compenstion Circuit

III. Design Methodology

The design of the proposed CMOS operational amplifier is based on optimizing the trade-off between noise performance, power consumption, voltage gain, bandwidth and stability. Since the amplifier is targeted for audio signal processing applications, special consideration is given to reducing input-referred noise while maintaining sufficient dynamic performance under a low supply voltage.

A. Noise Optimization Considerations

The major noise sources in CMOS operational amplifiers are thermal noise and flicker noise. Thermal noise is mainly generated due to the random movement of charge carriers in the MOS channel and is directly related to the transistor transconductance. The input-referred thermal noise voltage is inversely proportional to the transconductance. [5]

$$\overline{v_{n,thermal}^2} \propto \frac{1}{g_m} \tag{1}$$

where g_m is the transistor transconductance. Increasing g_m reduces the thermal noise contribution, but it requires higher bias current, creating a trade-off between noise and power consumption. [5]

Flicker noise is caused by charge trapping effects at the oxide-semiconductor interface and becomes dominant at low frequencies.

$$\overline{v_{n,flicker}^2} \propto \frac{1}{C_{ox}WLf} \tag{2}$$

where W and L are the transistor dimensions and f is the operating frequency.

Since audio applications involve low-frequency signals, reducing flicker noise is important. The input differential pair contributes significantly to the total input-referred noise because its noise is directly amplified by subsequent stages. Therefore, the input devices are designed with larger channel dimensions to increase transconductance and reduce both thermal and flicker noise contributions. The NMOS differential pair is selected due to its higher carrier mobility, providing better transconductance efficiency under the given power constraint.

B. Input Stage Design

The input stage devices are sized considering the trade-off between input-referred noise, transconductance and parasitic capacitance. Since the input devices have the highest contribution to the overall amplifier noise, larger channel dimensions are selected to reduce flicker noise while maintaining sufficient transconductance.

The NMOS input transistors are designed with W/L = 450 μm/4 μm to achieve higher transconductance efficiency under the available power budget. The PMOS load devices are sized at W/L = 600 μm/6 μm to increase output resistance and improve the first-stage gain. The tail current source transistor uses W/L = 35 μm/800 nm and is biased through the voltage generated by the beta multiplier circuit.

The selected device dimensions provide a transconductance of 1.7 mS for the input pair, providing a balance between noise performance, gain and power consumption.

C. Second Gain Stage Design

The second stage device dimensions are selected to achieve the required voltage gain and output operating point while minimizing the effect of parasitic capacitances on bandwidth and stability.

The PMOS gain transistor is sized with W/L = 150 μm/550 nm, while the NMOS current sink transistor uses W/L = 25 μm/500 nm. The selected sizing provides transconductances of 2.3 mS and 2.4 mS respectively, allowing sufficient gain contribution from the second stage.

The output bias condition is adjusted to approximately 0.9 V to maintain symmetrical output swing under the 1.8 V supply constraint. The device sizing also considers the effect of output capacitance on transient response and frequency compensation requirements.

D. Frequency Compensation Design

A Miller compensation network is implemented to improve the stability of the two-stage amplifier. The compensation capacitor is selected by considering the required phase margin and slew rate performance. A capacitor value of 3 pF is chosen to introduce a dominant pole while maintaining the required transient response.

The compensation resistor is selected based on the transconductance of the second gain stage. The resistor value of 417 Ω is used to control the compensation zero location and reduce the phase degradation caused by the right-half-plane zero. The selected compensation network provides a stable closed-loop response with ¿60° phase margin.

E. Bias Circuit Design

The bias circuit is designed to generate a stable bias voltage for the current sink transistors while maintaining proper transistor operating conditions under the 1.8 V supply constraint. The main design objective is to obtain a bias voltage of approximately 600 mV, providing suitable overdrive voltage for the NMOS current sinks while preserving sufficient voltage headroom.

IV. Simulation Results and Performance Analysis

The proposed operational amplifier is simulated using Cadence Virtuoso. DC, AC, transient and noise analyses are performed to evaluate the amplifier performance. The performance parameters including gain, bandwidth, stability, noise, transient response, linearity and power consumption are evaluated.

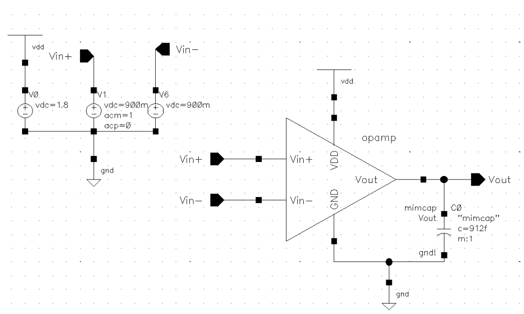

Fig. 6. Simulation Setup

All simulations were performed at a supply voltage of 1.8 V with a 1 pF capacitive load unless otherwise specified.

A. DC Gain, Bandwidth and Stability

The open-loop frequency response of the proposed operational amplifier is evaluated using AC analysis. The simulated results show a DC voltage gain of 91.4 dB.

The amplifier achieves a unity gain bandwidth of 46 MHz with a phase margin of 61°.

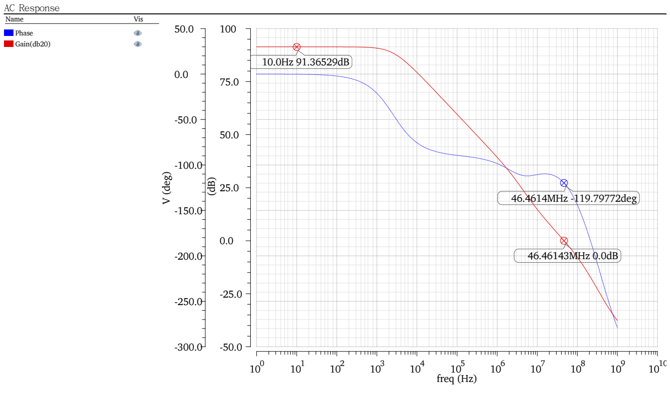

Fig. 7. AC response showing voltage gain, unity gain bandwidth and phase margin of the proposed op-amp.

The obtained phase margin indicates stable closed-loop operation with sufficient stability margin for feedback-based applications. The frequency compensation network effectively introduces a dominant pole and controls the phase response of the amplifier.

B. Noise Performance

The simulated input-referred noise is observed to be 8.6 nV/$\sqrt{Hz}$ at 1 kHz and reduces to 4.6 nV/$\sqrt{Hz}$ at 100 kHz. The reduction in noise density at higher frequencies is due to the decreasing contribution of flicker noise, which is dominant in the low-frequency region.

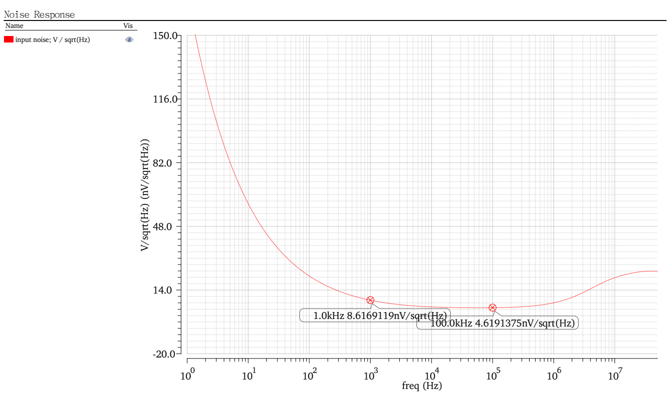

Fig. 8. Input-referred noise density of the proposed CMOS operational amplifier.

The obtained results indicate that the proposed amplifier provides low-noise operation suitable for audio front-end and analog signal conditioning applications.

C. Transient Response

The transient performance of the proposed operational amplifier is evaluated using slew rate and settling time analysis.

The simulated positive and negative slew rates are obtained as 14 V/μs and 14.5 V/μs respectively shown in Fig. 9. The closely matched rising and falling slew rates indicate symmetrical transient behavior.

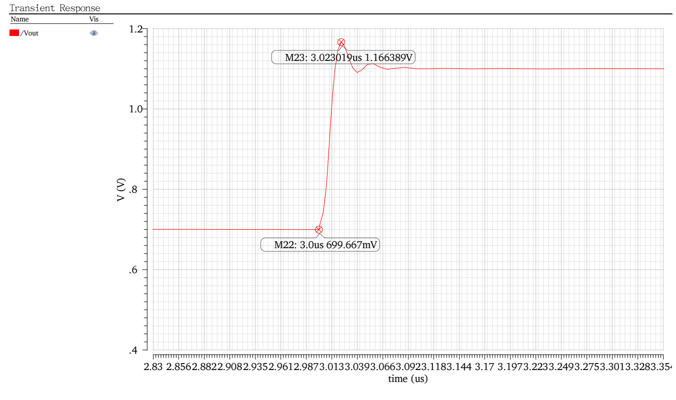

Fig. 9. Slew rate.

The settling behavior is evaluated for a 0.1% accuracy requirement. The amplifier achieves a settling time of 91 ns, demonstrating fast transient response suitable for audio signal processing applications. Shown in the Fig. 10.

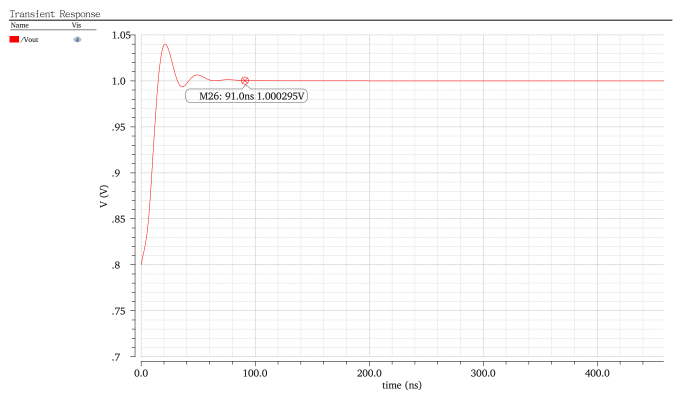

Fig. 10. Settling time.

D. Linearity and Distortion Performance

The linearity performance is evaluated using FFT analysis. The output spectrum shown in Fig. 11 demonstrates the harmonic components and the resulting THD of the amplifier. THD represents the amount of harmonic components introduced by the amplifier when processing a sinusoidal input signal.

The simulated amplifier achieves a total harmonic distortion of 0.00041%, corresponding to approximately -107.7 dB. The low distortion indicates that the amplifier maintains high signal fidelity with minimal nonlinear behavior.

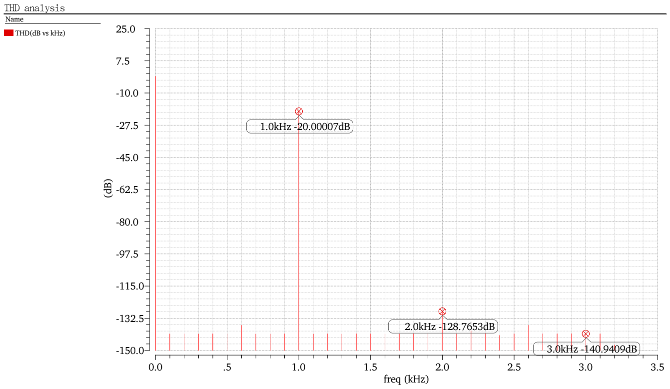

Fig. 11. FFT spectrum of the output signal showing harmonic distortion components.

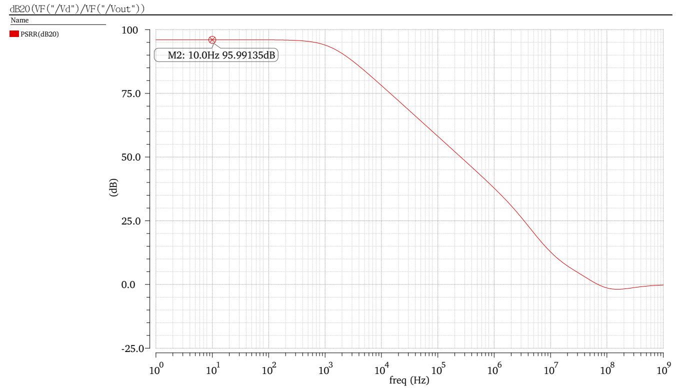

Fig. 13. PSRR performance.

E. Rejection Performance

The rejection capability of the operational amplifier is evaluated through common-mode rejection ratio (CMRR), power supply rejection ratio (PSRR) and temperature variation analysis.

The simulated CMRR of the op-amp is obtained as 93.5 dB as shown in the Fig. 12, indicating effective suppression of common-mode input signals. The high CMRR improves the accuracy of differential signal amplification by reducing unwanted common-mode interference.

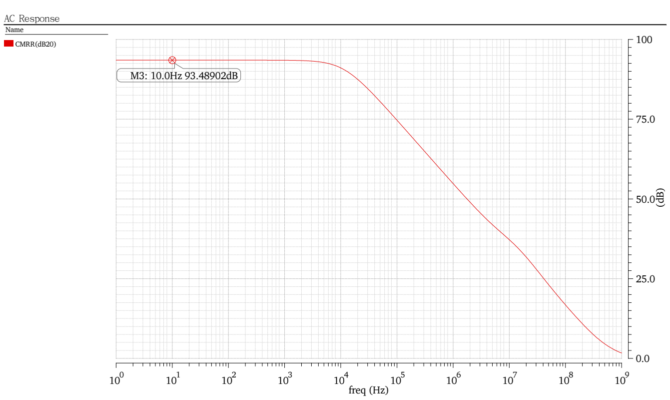

Fig. 12. CMRR performance.

The amplifier achieves a PSRR of 96 dB, demonstrating good immunity against supply voltage fluctuations shown in the Fig. 13 .

The temperature sweep analysis is performed to verify the stability of amplifier performance across different operating temperatures. The obtained results show stable behavior with minimal variation in the amplifier response, indicating reliable operation under practical conditions.

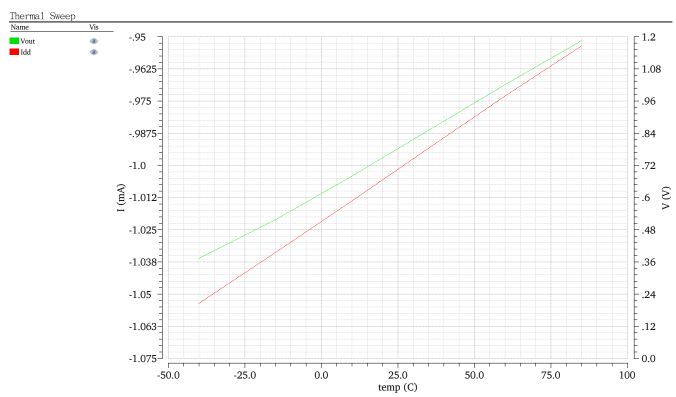

Fig. 14. Temperature variation analysis.

F. Output Swing and Power Consumption

The output swing capability of the proposed operational amplifier is evaluated under the 1.8 V supply voltage constraint. The amplifier achieves an output voltage range from 0 V to 1.74 V, providing a wide output swing close to the supply rails shown in the Fig. 14. This allows the amplifier to process larger signal amplitudes while operating at a low supply voltage.

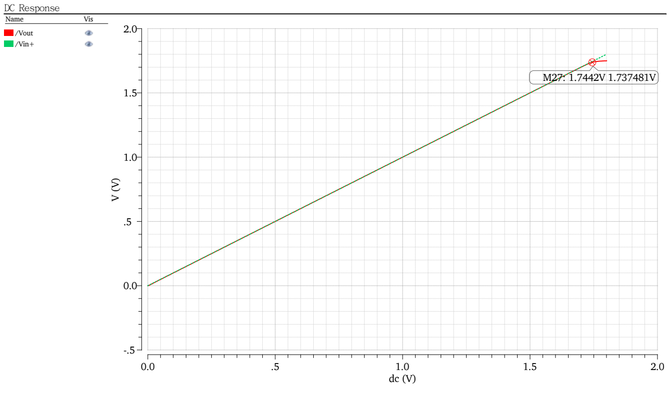

Fig. 15. Output voltage swing.

The power consumption is evaluated by measuring the total current drawn from the supply under normal operating conditions. The proposed design consumes approximately 0.92 mW of power. The low-power operation is achieved through proper bias current selection and transistor sizing while maintaining the required gain, bandwidth and noise performance.

V. Performance Comparison

The proposed op-amp is compared with two commercially available CMOS operational amplifiers selected from different performance categories. The TLV9062 is considered as a low-power general-purpose CMOS op-amp benchmark, while the OPA320 represents a higher performance low-noise CMOS amplifier suitable for precision analog signal processing applications. The comparison includes key parameters such as gain, bandwidth, noise performance, slew rate, distortion and power consumption.

TABLE I
Performance comparison of the proposed op-amp with commercial CMOS op-amps

Parameter	Proposed Op-Amp	TLV9062 [3]	OPA320 [4]
Technology	CMOS(180nM)	CMOS	CMOS
Supply Voltage	1.8 V	1.8-5.5 V	1.8-5.5 V
DC Gain	91.4 dB	∼100 dB	∼100 dB
UGB	46 MHz	10 MHz	20 MHz
PM (R_l=10k)	72°	55°	47°
Noise @ 1 kHz	8.6 nV/$\sqrt{Hz}$	16 nV/$\sqrt{Hz}$	8.5 nV/$\sqrt{Hz}$
Broadband Noise	4.6 nV/$\sqrt{Hz}$	10nV/$\sqrt{Hz}$	7nV/$\sqrt{Hz}$
Slew Rate	14 V/μs	6.5 V/μs	10 V/μs
Settling time(0.1%)	91ns(1pF)	500ns(100pF)	250ns(50pF)
THD	0.00041%	0.0008%	0.0005%
CMRR	93.5 dB	103 dB	114 dB
PSRR	96 dB	100dB	106dB
Power	0.92 mW	∼1 mW	∼1.45 mW

VI. Conclusion

A low-power, low-noise, wide-bandwidth two-stage CMOS operational amplifier for low-voltage audio signal processing applications has been designed and simulated. The proposed amplifier operates at 1.8 V supply voltage with a power consumption of 0.92 mW while achieving a DC gain of 91.4 dB, unity gain bandwidth of 46 MHz and phase margin of 61°.

The amplifier demonstrates low input-referred noise, high linearity with 0.00041% THD and suitable transient performance, making it applicable for noise-sensitive analog signal conditioning circuits. Future work includes layout implementation, parasitic extraction and PVT analysis for practical IC validation.

References

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